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Multiple-vibrational-mode model for fiber-optic Raman gain spectrum and response function

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We describe a model for the stimulated Raman gain spectrum and the Raman response function in silica fiber using multiple vibrational modes. We base the model on previous spectroscopic data [J. Opt. Soc. Am. B **1**, 652 (1984) and J. Opt. Soc. Am. B **6**, 1159 (1989)] and extend an earlier proposed model [Appl. Opt. **21**, 359 (1982)] by making use of an inhomogeneous distribution of damped oscillators. The model provides a simple analytical expression for the Raman response function and an expression for the Raman gain spectrum that is easy to evaluate numerically. © 2002 Optical Society of America

OCIS codes: 190.5650, 190.5890, 060.2320, 060.4370.

1. INTRODUCTION

In this paper we describe a new model for the stimulated Raman gain spectrum and a simple analytic expression for the Raman response function that leads to a good fit with experimental data. This model is an extension of the model proposed by Walrafen and Krishnan¹ and makes use of the spectroscopic data obtained by Stolen *et al.*^{2,3}

A numerically implementable model of the Raman gain spectrum and response function is useful for accurate and robust simulations of long-haul telecommunications systems including Raman amplification. Because the Raman response of fused silica is not instantaneous, it is necessary to model the Raman impulse response for numerical simulations of pulses that have short rise times or high peak power, such as solitons. A simple wavelength-dependent Raman gain can be used only if the time scale on which the optical intensity of the signal changes significantly is long compared with the Raman response time, which is of the order of 1 ps. Previous research⁴ has shown that depletion of a counterpropagating pump can occur on picosecond time scales, which are comparable with the Raman response time and therefore require incorporation of the transient Raman response.

We developed this model in the course of looking into the possibility of spectral hole burning in the Raman gain spectrum (Fig. 1). This investigation led to a quantum-mechanical model of the Raman interaction, which in turn led to the semiclassical model of the Raman gain spectrum and response function reported in this paper.

The principal results presented in this paper are summarized in Eqs. (9) and (11) in Section 3. Equation (9) is an analytical, easily evaluated expression for the Raman response function. Equation (11), which summarizes our model for the Raman gain function, can easily be evaluated by numerical integration. These results extend Walrafen and Krishnan's¹ Gaussian model to an intermediate-broadening model, which substantially im-

proves the agreement of the model gain and response functions with experiment.

2. PREVIOUS RESEARCH

We give a brief summary of the context of our model for the Raman gain and response function to clarify the notation that we use in this paper.

The standard approach employed when one is modeling the physical, or pulse-propagation, layer of a fiber-optic communication system is to solve the fiber paraxial wave equation using one of several different methods. The fiber paraxial wave equation (including the standard Taylor-series expansion of the dispersion) for the slowly varying field envelope of the Stokes or signal field $\mathcal{E}_S(z, t)$ can be written in the form

$$\left(\frac{\partial}{\partial z} + \beta_1 \frac{\partial}{\partial t} + i \frac{1}{2} \beta_2 \frac{\partial^2}{\partial t^2} - \frac{1}{6} \beta_3 \frac{\partial^3}{\partial t^3} \right) \mathcal{E}_S(z, t) = -\frac{\alpha}{2} \mathcal{E}_S(z, t) + \frac{2\pi i \omega_0^2}{\beta_0 c^2} \mathcal{P}_S(z, t), \quad (1)$$

where the dispersion parameters are

$$\beta_j = \left. \frac{d^j \beta}{d\omega^j} \right|_{\omega=\omega_0}, \quad (2)$$

α is the linear attenuation, β is the propagation constant (determined when we solve the fiber mode eigenvalue-eigenfunction problem), and $\mathcal{P}(z, t)$ is the nonlinear polarization, which acts as a source term. The nonlinear polarization includes both an instantaneous (electronic) term and a time-delayed (vibrational) term,

$$\mathcal{P}_S = \mathcal{P}_{S,i} + \mathcal{P}_{S,R}. \quad (3)$$

The vibrational term $\mathcal{P}_{S,R}$ is the source of the Raman-shifted light. The instantaneous term can be taken to be of the form

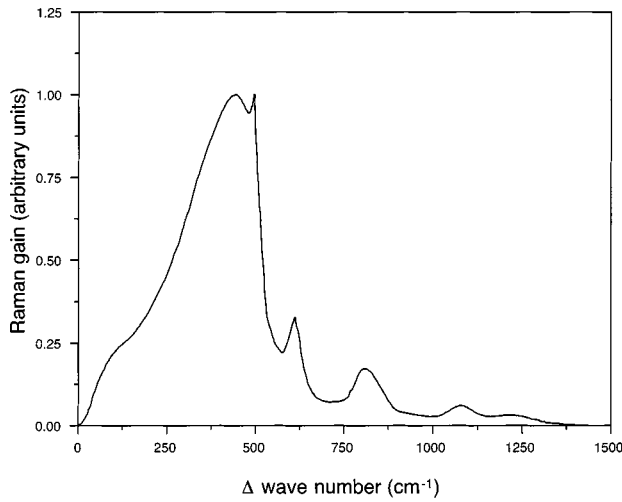


Fig. 1. Raman gain spectrum of silica fiber, after Stolen *et al.*³

$$\mathcal{P}_{S,i} \propto (1 - f_R) \mathcal{E}_S[|\mathcal{E}_S|^2 + 2|\mathcal{E}_L|^2], \quad (4)$$

where the Raman fraction $f_R = 0.18$, which can be derived from the maximum Raman gain, represents the fraction of the polarization contributed by the Raman effect. The Raman contribution to the nonlinear polarization is

$$\begin{aligned} \mathcal{P}_{S,R}(t) &\propto i f_R \mathcal{E}_L(z, t) \\ &\times \int_{-\infty}^{\infty} h_R(t - t''', 0) \exp[-i(\omega_L - \omega_S)(t - t''')] \\ &\times \mathcal{E}_L^*(z, t''') \mathcal{E}_S(z, t''') dt''', \end{aligned} \quad (5)$$

where $\mathcal{E}_L$ is the laser or pump-field envelope. As can be seen, the Raman polarization is a product of the laser electric field envelope and a convolution of an oscillator impulse response function h_R , a frequency-matching term, and a product of the laser and Stokes electric field envelopes that comes from the force that drives the oscillator(s). Our model provides an analytic expression for this response function and its Fourier transform, the imaginary part of which is the Raman spectrum. One can write a similar set of equations for the laser field envelope $\mathcal{E}_L$.

It is apparent that the experimental Raman gain spectrum shown in Fig. 1 cannot reasonably be modeled as a singly peaked function, and especially not as a triangle⁵ or as a single Lorentzian or Gaussian. A cursory examination of Fig. 1, in which there are at least seven humps or peaks, suggests that many vibrational modes contribute to the Raman gain in silica fiber. From the fact that several of the seven peaks in Fig. 1 are quite asymmetrical, one can estimate crudely that a total of 10–15 vibrational modes contribute to the Raman spectrum in silica fiber. Walrafen and Krishnan,¹ for example, showed ten peaks (up to 830 cm^{-1}) in their synthesized Raman gain spectrum, and included 13 peaks with measurable amplitudes in an accompanying table.

The response function that can be seen in Fig. 2 contains a great deal of structure that is not possible to model as a single damped oscillation. In fact, the response function resembles a superposition of oscillatory functions that are interfering constructively and destruc-

tively. Especially noteworthy are the almost total decay around 350 fs in addition to the recurrence after approximately 500 fs and the flat minimum (phase change) around 160 fs, neither of which is consistent with a single damped sinusoid.

Two models of the Raman response function are widely used: an instantaneous model, in which $h_R(t) = \delta(t)$, and the single-damped-harmonic-oscillator approximation that is attributable to Blow and Wood,⁶ in which

$$h_R(t) \propto \exp(-t/\tau_2) \sin(t/\tau_1). \quad (6)$$

In relation (6) $\tau_1 = 1/\omega_v$ and $\tau_2 = 1/\gamma$ are adjustable parameters used to fit the actual Raman gain spectrum and are taken to be $\tau_1 = 12.2$ fs and $\tau_2 = 32$ fs. This leads to an impulse response function of the form shown in Fig. 3, where it can be seen that the initial behavior (up to 100 fs) is an adequate match for the measured response function [Stolen *et al.*,³ Fig. 3(e)]. However, past 100–150 fs there is little or no similarity between the measured response shown in Fig. 2 and the single-damped-oscillator model shown in Fig. 3 based on relation (6).

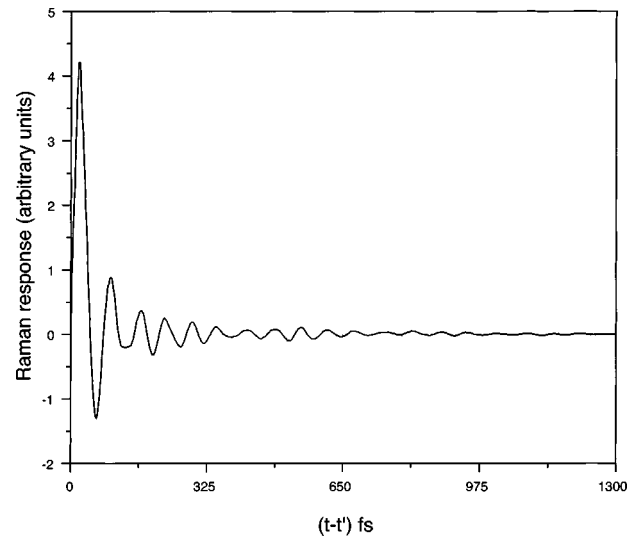


Fig. 2. Raman response function of silica fiber, after Stolen *et al.*³

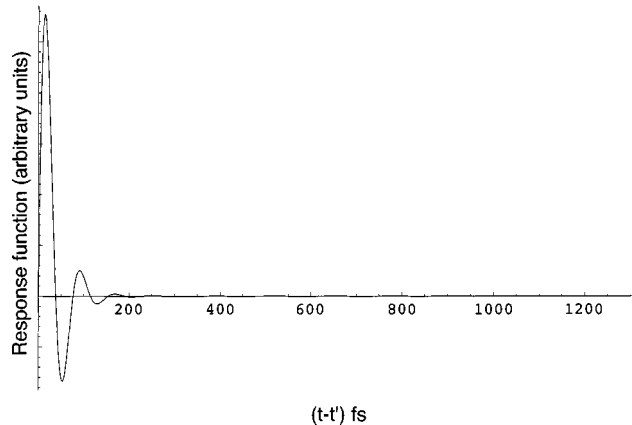


Fig. 3. Response function in the single-damped-oscillator model [relation (6)].

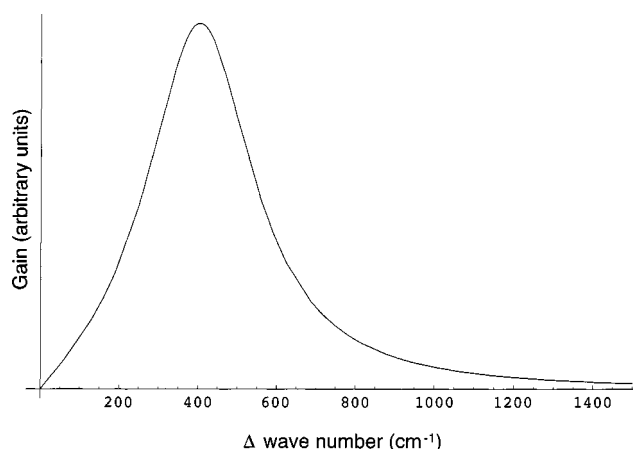


Fig. 4. Gain spectrum in the single-damped-oscillator model [relation (8)].

The Raman gain spectrum g_R can be obtained from the Raman response function by the relation [Agrawal,⁷ Eq. (2.3.42)]

$$g_R(\omega) = \frac{\omega_0}{cn_0} f_R \chi^{(3)} \text{Im}[\tilde{h}_R(\omega)], \quad (7)$$

where the transfer function $\tilde{h}_R(\omega)$ is the Fourier transform of the impulse response function. In Eq. (7), ω_0 is the center frequency, c is the speed of light in vacuum, n_0 is the index of refraction of the silica fiber, and $\chi^{(3)}$ is the third-order susceptibility.

Applying the transformation expressed in Eq. (7) to the single-damped-oscillator expression used by Blow and Wood,⁶ our Eq. (6) yields

$$g_R(\omega) \propto \frac{2\omega\gamma}{(\omega^2 - \omega_v^2)^2 + (2\omega\gamma)^2}. \quad (8)$$

Extraneous constants were removed to emphasize the functional form, which can be seen in Fig. 4.

Although Figs. 3 and 4 illustrate the overall gross structure of the Raman gain spectrum and the Raman response function, all the fine structure is absent. It is precisely this fine structure that we model in this paper.

Blow and Wood's⁶ approach was based on assuming that the silica molecules in the fiber truly behave as simple harmonic oscillators. This approach leads to a Lorentzian gain spectrum and a damped-forced-harmonic-oscillator response function. However, optical fiber is not just a collection of independent silica molecules that all behave in the same manner, but is a random arrangement of silicon and oxygen atoms with an ensemble of different internal structures consistent with the vitreous nature of the material.

Many authors, including Bell and Dean,⁸ Walrafen and Krishnan,¹ and Revesz and Walrafen,⁹ have attempted to assign specific vibrational modes to the peaks observed in the Raman spectrum of silica. After considering the individual features of the Raman spectrum of silica, Walrafen and Krishnan (Table 1)¹ proposed 13 Gaussians with various widths and amplitudes, centered at various frequency offsets up to 1180 cm^{-1} , along with modes at higher frequencies for which no amplitudes were given, for use in modeling the experimental Raman gain profile.

Figure 5 and Table 1 show the 13 Gaussians used in our intermediate-broadening model. In Walrafen and Krishnan's model each Gaussian corresponds to a different vibrational mode in fused silica. A Gaussian profile would be expected for a superposition of a large number of sharper Lorentzian spectral peaks, each of which corresponds to the vibrational frequency of an individual atomic group that has been randomly perturbed by neighboring atoms. The central limit theorem implies that, in the limit of a large number of random perturbations, the vibrational frequency of a particular mode should be a Gaussian random variable with a variance that depends on the perturbing forces. This is the limiting case known in quantum electronics as purely inhomogeneous broadening.

In Walrafen and Krishnan's paper¹ and in Revesz and Walrafen,⁹ efforts were made to assign some of their tabulated 13 modes to specific SiO_2 vibrational modes. It may be useful to note that Walrafen and Krishnan's spectrum¹ is not corrected with the thermal ground-state population factor as in Stolen *et al.*² This discrepancy is the cause of the difference between Walrafen's and Stolen's spectra at low-frequency shifts.

However, Walrafen failed to take into account the fact that the widths of the inhomogeneous distributions of vibrational frequencies may not be large enough compared with the widths of the Lorentzian spectra of individual vibrational groups to make the spectrum of each mode a pure Gaussian, corresponding to purely inhomogeneous broadening.

It is well known that the spectrum of an ensemble of damped harmonic oscillators, the resonant frequencies of which have a Gaussian distribution, is the convolution of a Gaussian and a Lorentzian.¹⁰ The same is true of an inhomogeneously distributed ensemble of atoms.¹¹

In Appendix A we illustrate the evaluation of the Raman response function and the Raman gain spectrum in a fully inhomogeneous model, as in Walrafen's model. We do the same in Appendix B in the purely homogeneous limit. The result is that the parameters in the fully inhomogeneous model and the fully homogeneous model cannot be adjusted to give simultaneously good fits to the Raman gain spectrum and the Raman response function.

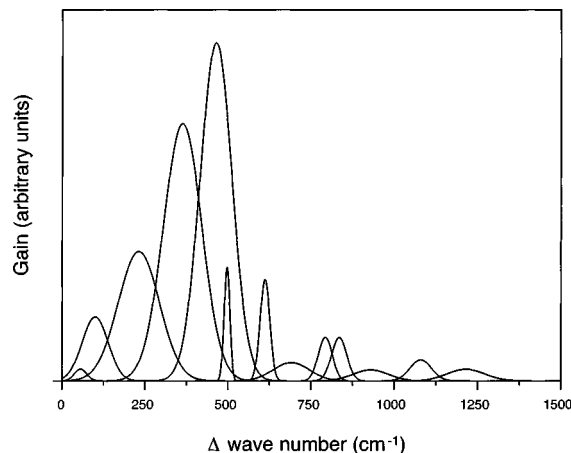


Fig. 5. Gaussians used in the intermediate-broadening model of the Raman gain spectrum.

Table 1. Values of the Parameters Used in the Intermediate-Broadening Model^a

Mode Number <i>i</i>	Component Position (cm ⁻¹)	Peak Intensity <i>A_i</i>	Gaussian FWHM (cm ⁻¹)	Lorentzian FWHM (cm ⁻¹)
1	56.25	1.00	52.10	17.37
2	100.00	11.40	110.42	38.81
3	231.25	36.67	175.00	58.33
4	362.50	67.67	162.50	54.17
5	463.00	74.00	135.33	45.11
6	497.00	4.50	24.50	8.17
7	611.50	6.80	41.50	13.83
8	691.67	4.60	155.00	51.67
9	793.67	4.20	59.50	19.83
10	835.50	4.50	64.30	21.43
11	930.00	2.70	150.00	50.00
12	1080.00	3.10	91.00	30.33
13	1215.00	3.00	160.00	53.33

^a $A_i = A_i'/\omega_{v,i}$, $\Gamma_i = \pi c \times \text{Gaussian FWHM}$, $\gamma_i = \pi c \times \text{Lorentzian FWHM}$, and $\omega_{v,i} = 2\pi c \times \text{component position}$.

In Section 3 we describe an intermediate-broadening model using convolutions of Lorentzians with Gaussians, the parameters in which can be chosen such that the model's Raman gain spectrum and Raman response function both fit well with experiment.

3. INTERMEDIATE BROADENING

Figure 6 illustrates five members of an infinite set of Lorentzians that are convolved with a Gaussian. For the peak in the Raman spectrum at approximately 440 cm⁻¹, for example, which corresponds to the bending of an Si-O-Si dihedral angle, each Lorentzian peak would correspond to a different equilibrium value of the dihedral angle.⁹

Our intermediate-broadening model for the Raman response function for intermediate broadening is

$$h_R(t) = \sum_{i=1}^{13} \frac{A_i'}{\omega_{v,i}} \exp(-\gamma_i t) \exp(-\Gamma_i^2 t^2/4) \sin(\omega_{v,i} t) \theta(t), \quad (9)$$

where A_i' is the amplitude of the i th vibrational mode, $\omega_{v,i}$ is the center vibrational frequency for mode i , γ_i is the Lorentzian linewidth for mode i , Γ_i is the Gaussian linewidth for mode i , and $\theta(t)$ is the unit step function such that

$$\theta(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases} \quad (10)$$

Figure 7 shows our best fit of Eq. (9) to the experimental Raman response function. The initial sharp peak, the phase reversal around 160 fs, and the recurrence around 500 fs are all fit well.

The Raman gain spectrum, which is proportional to the imaginary part of the transfer function [the Fourier transform of the impulse response function given in Eq. (9)],

$$s(\omega) = \sum_{\ell=1}^{13} \frac{A_{\ell}'}{2\omega_{v,\ell}} \int_0^{\infty} \{\cos[(\omega_{v,\ell} - \omega)t] - \cos[(\omega_{v,\ell} + \omega)t]\} \exp(-\gamma_{\ell} t) \exp(-\Gamma_{\ell}^2 t^2/4) dt, \quad (11)$$

is shown in Fig. 8. Small differences from the experimental spectrum occur on the shoulders around 300 and 530 cm⁻¹ and in the valley around 480 cm⁻¹. Overall, the model fits the experimental data well.

To invert the Fourier transform, one also needs the real part of the transfer function,

$$r(\omega) = \sum_{\ell=1}^{13} \frac{A_{\ell}'}{2\omega_{v,\ell}} \int_0^{\infty} \{\sin[(\omega_{v,\ell} - \omega)t] + \sin[(\omega_{v,\ell} + \omega)t]\} \exp(-\gamma_{\ell} t) \exp(-\Gamma_{\ell}^2 t^2/4) dt, \quad (12)$$

which is shown in Fig. 9 for comparison with Stolen *et al.*, Fig. 1(b).³

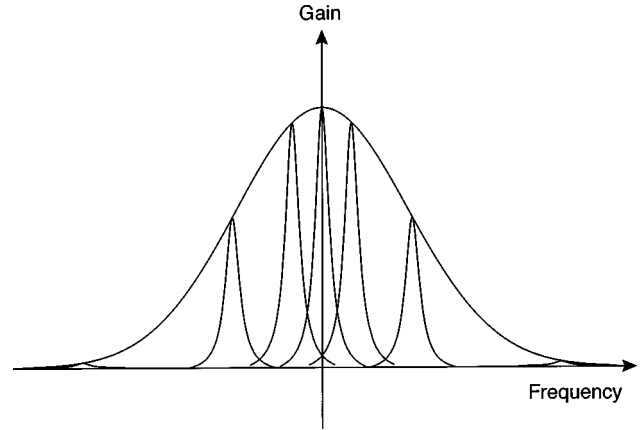


Fig. 6. Representation of the convolution of Lorentzians with a Gaussian.

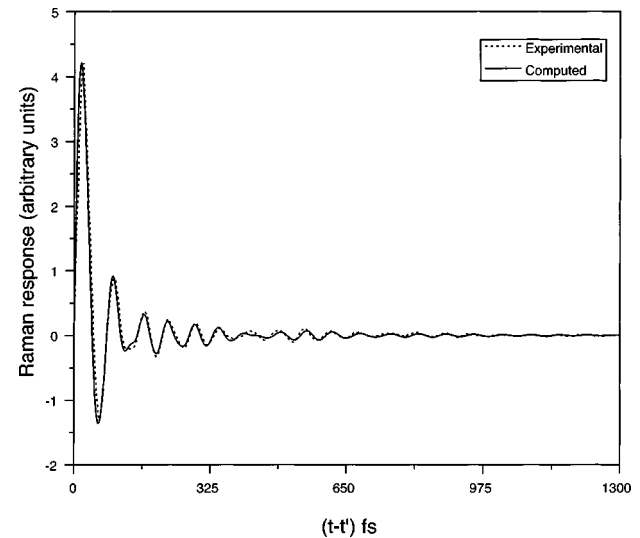


Fig. 7. Best fit achieved for the Raman response function in the intermediate-broadening model [Eq. (9)] and the data from Table 1.

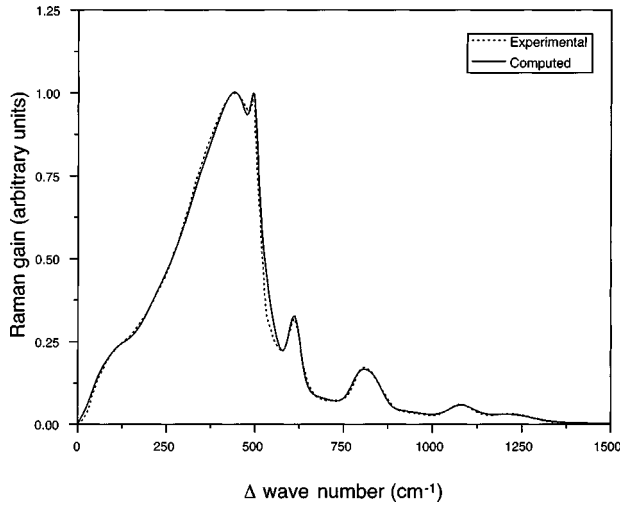


Fig. 8. Best fit achieved for the Raman gain spectrum $g_R(\omega) \propto s(\omega)$ with the intermediate-broadening model [Eq. (11)] and the data given in Table 1.

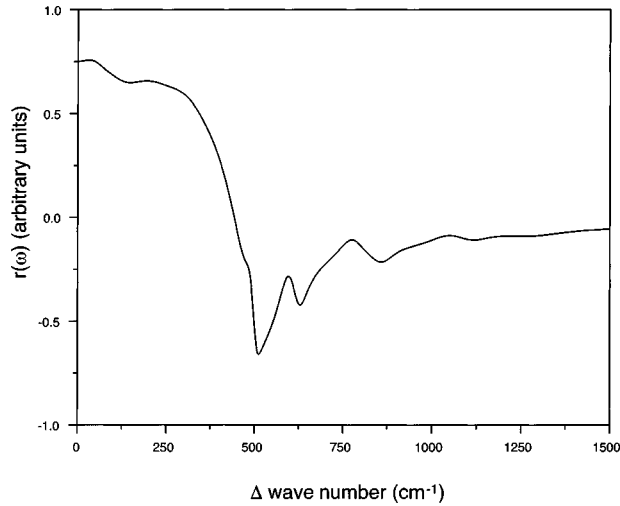


Fig. 9. Real component of the Raman transfer function in the intermediate-broadening model [Eq. (12)] and the data from Table 1.

As a result of the complex nonlinear coupling of the parameters of the model and the requirement of fitting both the response function and the spectrum with physically reasonable parameters, we found it easier to fit by hand rather than with a least-squares program. We attempted to stay as close as possible to Walrafen's and Krishnan's¹ vibrational frequencies $\omega_{v,i}$ and Gaussian linewidths Γ_i to minimize the number of free parameters. (Walrafen's and Krishnan's Table 1 is mislabeled. They tabulated full widths not half-widths.) Instead, we mainly adjusted the modal amplitudes A_i' . We chose the ratio of the Gaussian linewidth to the Lorentzian linewidth Γ_i/γ_i to be 3 so as to fit the response function and the spectrum with minimal modifications to Walrafen's and Krishnan's original parameters.

As can be seen in Figs. 7 and 8, the intermediate-broadening model produces a good fit for the Raman response function and Raman gain spectrum by use of the set of parameters given in Table 1. This fit is a great improvement over the purely inhomogeneous model or the

purely homogeneous model, which are discussed in Appendixes A and B, respectively.

4. DISCUSSION AND CONCLUSIONS

In this paper we give a numerically viable and accurate model for the Raman response function and gain spectrum. This model expands on the spectroscopic ideas presented in Walrafen and Krishnan¹ to account for the finite Lorentzian linewidth of the vibrational modes in silica fiber. However, we do not claim spectroscopic accuracy in our model. We simply report a combination of oscillator frequencies, linewidths, and line shapes that leads to substantially improved agreement with the experiment. Our research further improves on Walrafen's and Krishnan's research through a simultaneous fit of the Raman gain spectrum and the Raman response function.

The intermediate-broadening model has been used successfully to allow for a more physical coupling of the pump and signal fields than can be obtained in either the Blow and Wood⁶ or the instantaneous response function models to model laser-pulse propagation in long-haul telecommunication systems.

A possible disadvantage of this model is that the parameters given in Table 1 may not be the same for all fibers. However, it is our opinion that most single-mode silica fibers will have frequency and linewidth parameters that are close to those given in Table 1, although the amplitudes of some modes may vary from our values. Dopants may possibly give rise to additional weak modes. In fibers with complicated refractive-index profiles in the core, care should be exercised when Raman spectral measurements obtained at nontelecommunications wavelengths are applied to obtain parameters for our model, because the intensity distribution in the fiber core depends on the wavelength of the probe used to obtain the Raman spectrum.

APPENDIX A: PURELY INHOMOGENEOUSLY BROADENED OSCILLATORS

In the limit of purely inhomogeneous broadening ($\gamma_i \ll \Gamma_i$), the Raman gain spectrum is

$$g_R(\omega) \propto \sum_{i=1}^{13} \frac{A_i'}{2\omega_{v,i}} \exp[(\omega - \omega_{v,i})^2/\Gamma_i^2] \quad (\text{A1})$$

and the Raman response function is

$$h_R(t) = \sum_{i=1}^{13} \frac{A_i'}{\omega_{v,i}} \exp(-\Gamma_i^2 t^2/4) \sin(\omega_{v,i} t) \theta(t), \quad (\text{A2})$$

where θ is the unit step function,

$$\theta(t) = \begin{cases} 1, & \text{if } t \geq 0 \\ 0, & \text{if } t < 0 \end{cases} \quad (\text{A3})$$

The Raman gain spectrum in the purely inhomogeneous limit, by use of the parameters provided in Table 1, is shown in Fig. 10.

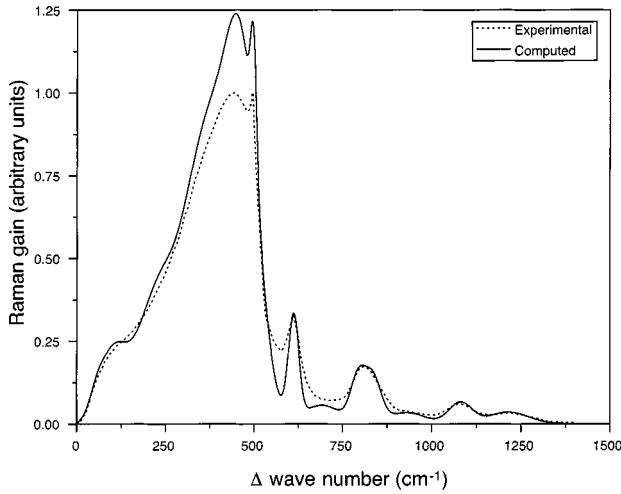


Fig. 10. Raman gain spectrum in the limit of purely inhomogeneous broadening.

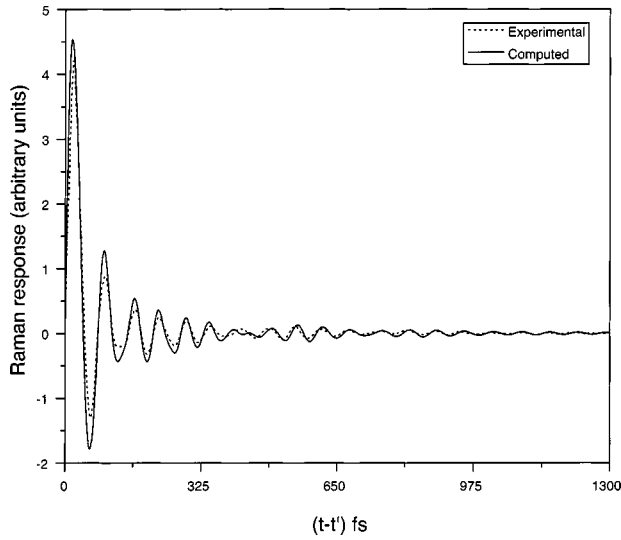


Fig. 11. Raman response function in the limit of purely inhomogeneous broadening.

From Fig. 10 it is evident that the purely inhomogeneous model expressed in relation (A1) fits the overall shape of the experimental spectrum only moderately well.

From Eq. (7), one can see that if the Raman gain is expressed as a sum of Gaussians in frequency, then the impulse response function should be a sum of products of Gaussians and sinusoids in time. As can be seen in Fig. 11, the response function in the model of purely inhomogeneous broadening fits the experiment fairly well, except at some of the more prominent peaks. Even by adjusting the modal amplitudes specifically for this model, we were unable to obtain as good a fit to the experimental response function and gain spectrum as in the intermediate-broadening model.

These observations suggest that each individual peak may not be sufficiently inhomogeneous to fit well in a Gaussian model.

APPENDIX B: PURELY HOMOGENEOUSLY BROADENED OSCILLATORS

A classical simple harmonic-oscillator model results in a Raman gain spectrum that is a sum of Lorentzians, which

leads to a better fit for the Raman response function but a poorer fit for the spectrum than the purely inhomogeneous model in relation (A1).

For a Raman gain spectrum that is a sum of Lorentzians, the Fourier-transform relation between the impulse response function and the Raman gain profile, Eq. (7), implies that the vibrational (Raman) impulse response function is a linear combination of exponentials multiplied by sinusoids. An approximate impulse response function in the limit of purely homogeneous broadening is

$$h_R(t) = \sum_{i=1}^{13} \frac{A_i'}{\omega_{v,i}} \sin(\omega_{v,i}t) \exp(-\gamma_i t), \quad (\text{B1})$$

as shown in Fig. 12. The response function computed in this model accounts reasonably well for the phase reversal around 160 fs, but is a poor fit to the recurrence after 500 fs. An approximate expression for the Raman gain spectrum in the same limit is

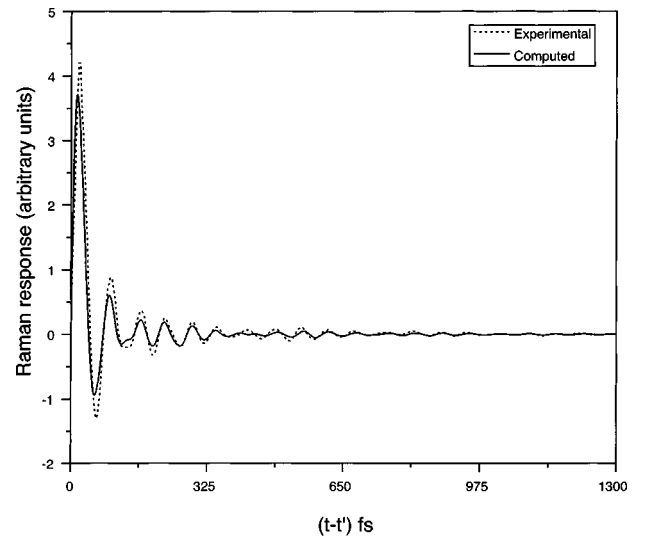


Fig. 12. Raman response function in the limit of purely homogeneous broadening ($\Gamma_i = 0$).

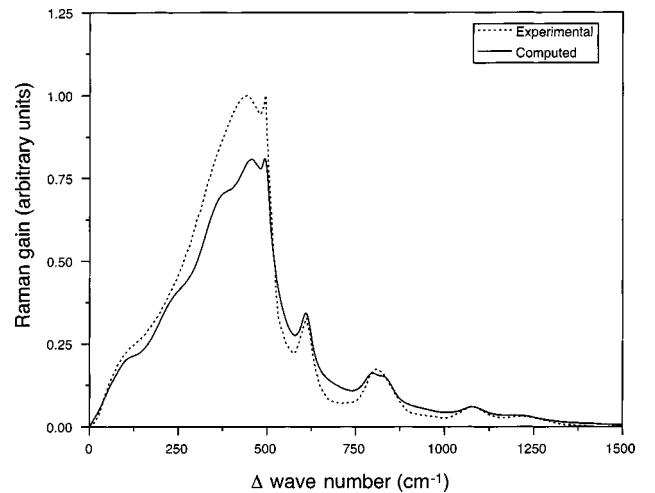


Fig. 13. Raman gain spectrum in the limit of purely homogeneous broadening ($\Gamma_i = 0$).

$$g_r(\omega) \propto \sum_{i=1}^{13} \frac{A_i'}{2\omega_{v,i}} \times \frac{2\omega_{v,i}\gamma_i\omega}{\gamma_i^4 + (\omega^2 - \omega_{v,i}^2)^2 + 2\gamma_i^2(\omega^2 + \omega_{v,i}^2)}, \quad (\text{B2})$$

as shown in Fig. 13. The purely homogeneous model does not give a good fit to the low-frequency shoulder from 0 to 400 cm^{-1} , or to the valleys at higher frequencies.

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REFERENCES

1. G. E. Walrafen and P. N. Krishnan, "Model analysis of the Raman spectrum from fused silica optical fibers," *Appl. Opt.* **21**, 359–360 (1982).
2. R. H. Stolen, C. Lee, and R. K. Jain, "Development of the stimulated Raman spectrum in single-mode silica fibers," *J. Opt. Soc. Am. B* **1**, 652–657 (1984).
3. R. H. Stolen, J. P. Gordon, W. J. Tomlinson, and H. A. Haus, "Raman response function of silica-core fibers," *J. Opt. Soc. Am. B* **6**, 1159–1166 (1989).
4. D. Hollenbeck, "Dynamics of a fiberoptic Raman amplifier," Ph.D. dissertation (University of Texas at Dallas, Dallas, Tex., 2000).
5. A. R. Chraplyvy, "Optical power limits in multichannel wavelength-division-multiplexed systems due to stimulated Raman scattering," *Electron. Lett.* **20**, 58–59 (1984).
6. K. J. Blow and D. Wood, "Theoretical description of transient stimulated Raman scattering in optical fibers," *IEEE J. Quantum Electron.* **25**, 2665–2673 (1989).
7. G. P. Agrawal, *Nonlinear Fiber Optics* (Academic, San Diego, Calif., 1997).
8. R. J. Bell and P. Dean, "Atomic vibrations in vitreous silica," *Discuss. Faraday Soc.* **50**, 55–61 (1970).
9. A. G. Revesz and G. E. Walrafen, "Structural interpretations for some Raman lines from vitreous silica," *J. Non-Cryst. Solids* **54**, 323–333 (1983).
10. A. C. G. Mitchell and M. W. Zemansky, *Resonance Radiation and Excited Atoms* (Cambridge U. Press, New York, 1971), p. 100.
11. A. Içsevci and W. E. Lamb, Jr., "Propagation of light pulses in a laser amplifier," *Phys. Rev.* **185**, 517–545 (1969).